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Technology - European Semiconductors | Europe

# Global Technology Webcast – ASML Management Roadshow, Memory Innovation and Apple AI

Join technology analysts Shawn Kim, Daniel Yen, Dylan Liu and Amelia Scicluna to discuss key takeaways from our roadshow with ASML CEO and CFO, Apple AI to China as a key catalyst for AAPL and innovating the next generation of memory. Register [here](#).

|                                         |                                   |                  |
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| Europe                               |         |
| Industry View                        | In-Line |

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OVERVIEW

Global Technology Webcast  
ASML Management Roadshow, Memory Innovation and Apple AI



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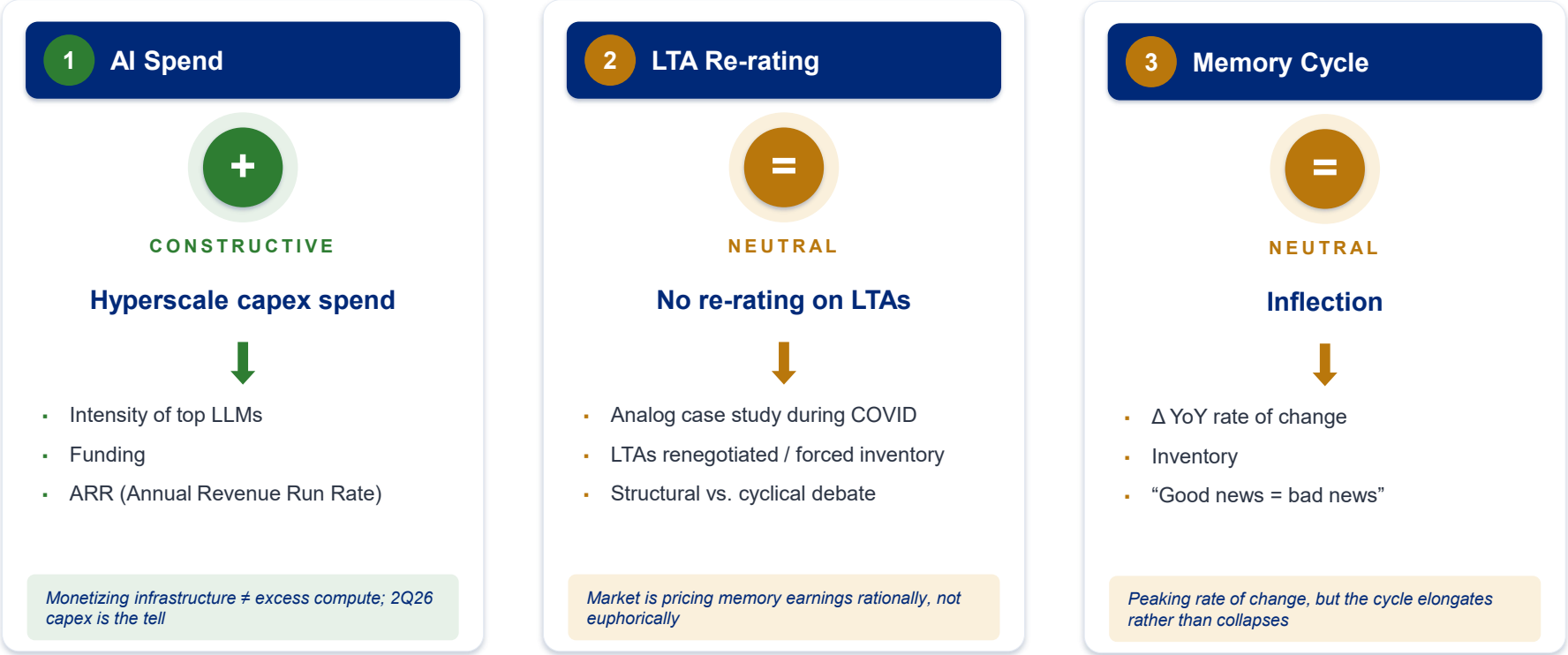
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Shawn Kim

Head of Asia and Europe Technology Research

Memory – Changing Tides: Three Debates Framing the Cycle

How we frame the memory debate for investors – where the money is being spent, how the market prices LTAs, and where the cycle now sits.



Source: Morgan Stanley Research

Kimi K3: China’s 2.8T open-weight model resets the AI frontier

Moonshot AI’s 2.8T-parameter open model (16 Jul) benchmarks just behind Fable 5 and GPT-5.6 Sol – the first Chinese open release competing on capability, not just price.

2.8T

total parameters  
(MoE: 16 of 896 experts active)

1M

token context window; native vision

\$3 / \$15

per M tokens (input / output)

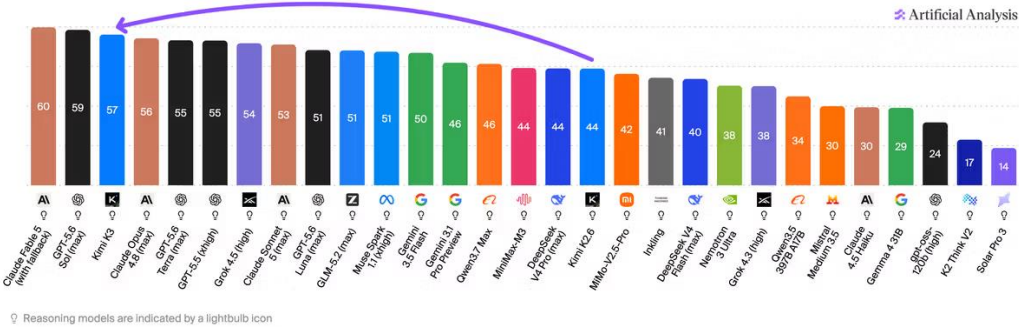
#3

frontier rank; #1 open-weight

Kimi K3 vs. the AI frontier

Artificial Analysis Intelligence Index

Artificial Analysis Intelligence Index v4.1 incorporates 9 evaluations: GDPval-AA v2, <sup>2</sup>-Banking, Terminal-Bench v2.1, SciCode, Humanity’s Last Exam, GPQA Diamond, CritPT, AA-Omniscience, AA-LCR



Reasoning models are indicated by a lightbulb icon

What’s new

- Architecture: Kimi Delta Attention + Attention Residuals; sharply improved scaling efficiency vs K2
- Open weights due 27 Jul; undercuts Fable 5 / GPT-5.6 Sol sharply on price
- Backed by Alibaba; launched ahead of the 2026 World AI Conference (Shanghai)
- Caveat: independent hallucination rate rose to ~51% – keep verification in the loop

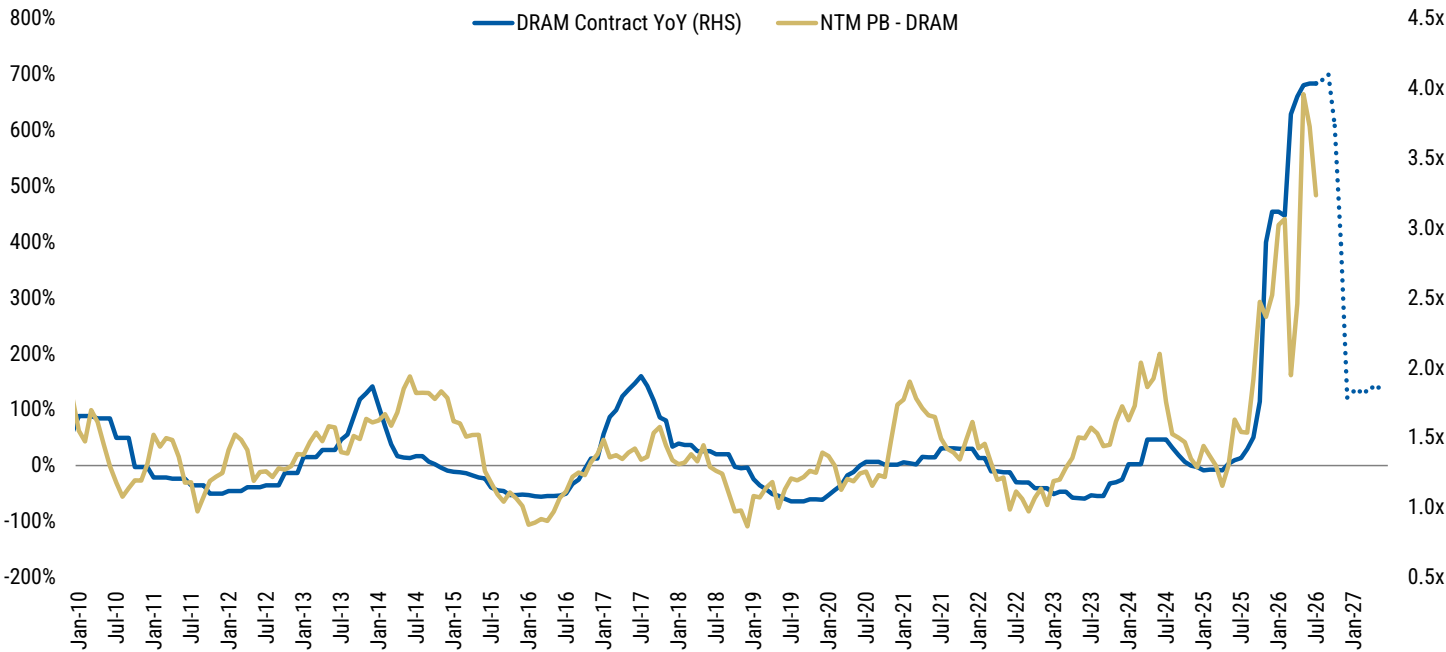
Source: Moonshot AI; Artificial Analysis; VentureBeat, Tom’s Hardware. \*with tools.

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Memory pricing likely to peak in 4Q26

DRAM contract price YoY is rolling over from cyclical peaks, while valuation (NTM P/B) has yet to re-rate – consistent with pricing peaking around 4Q26.



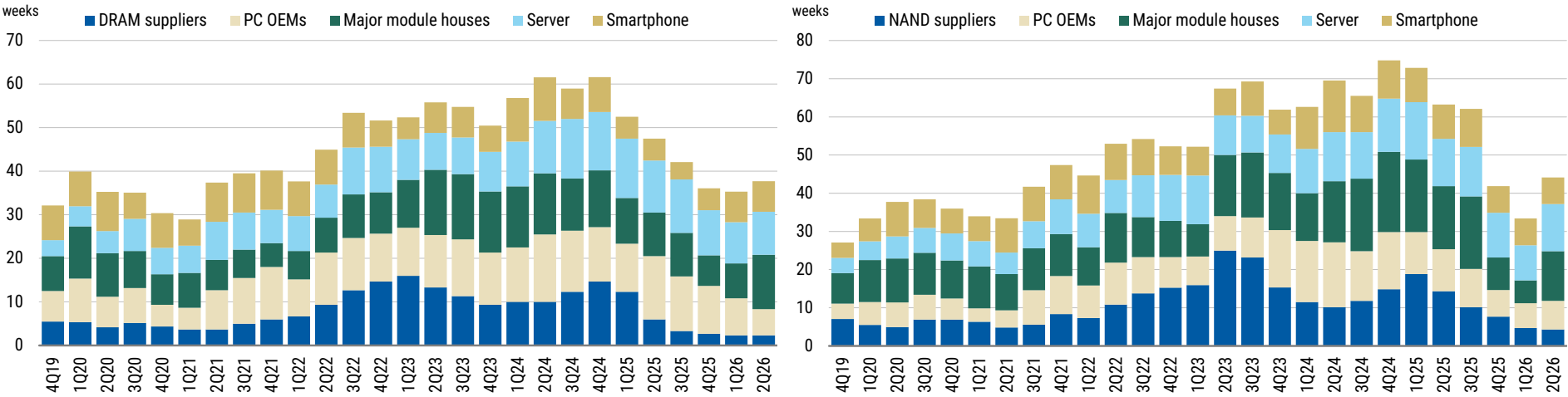
Source: Trendforce, Factset, Morgan Stanley Research forecasts (dotted line)

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DRAM & NAND Inventory picked up in 2Q26

... primarily driven by module houses



Source: Trendforce, Morgan Stanley Research

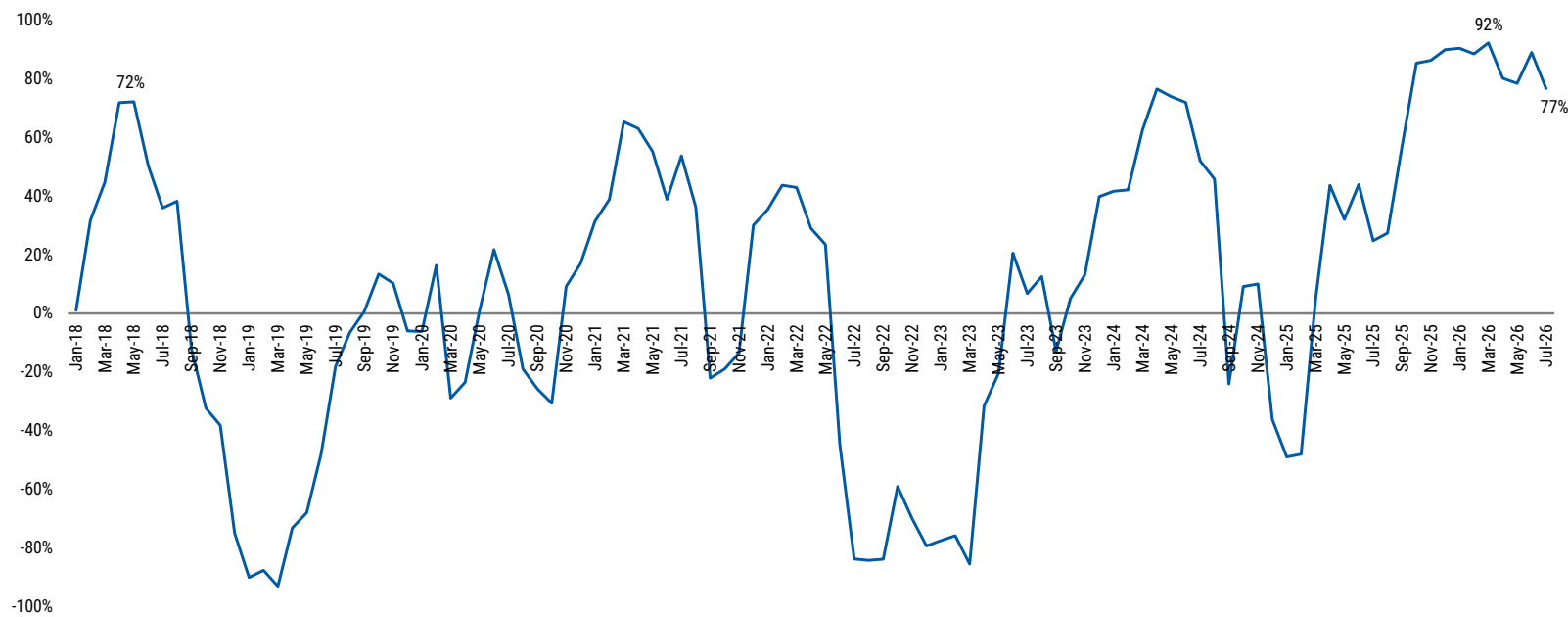


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Earnings revision rate losing momentum

Net upward earnings revisions have rolled over from a 92% peak to 77% – still positive, but an early sign the upgrade cycle is losing steam.



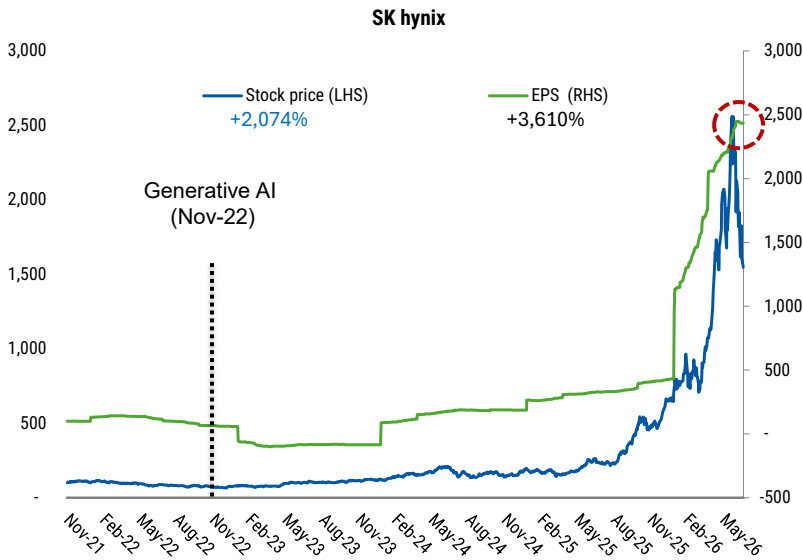
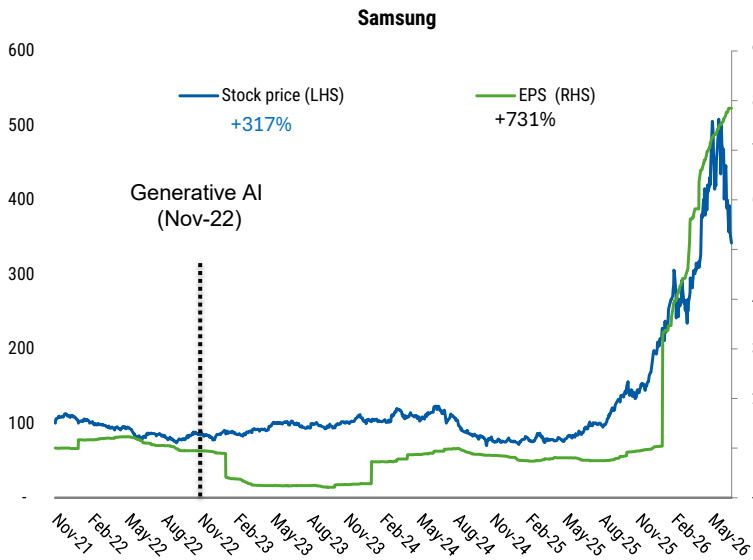
Source: Factset, Morgan Stanley Research estimates

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EPS vs. Share price

1yr forward EPS has come off recently for SK Hynix

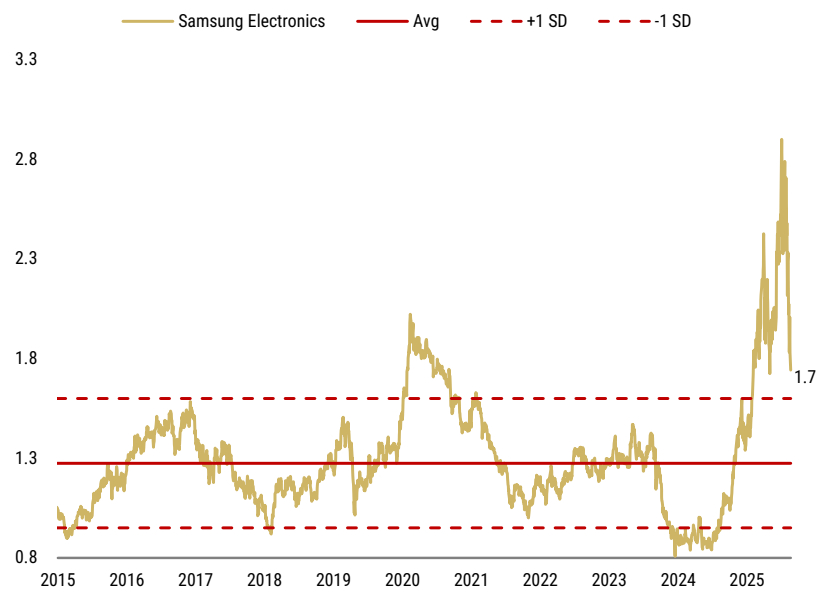


Source: Factset, Morgan Stanley Research

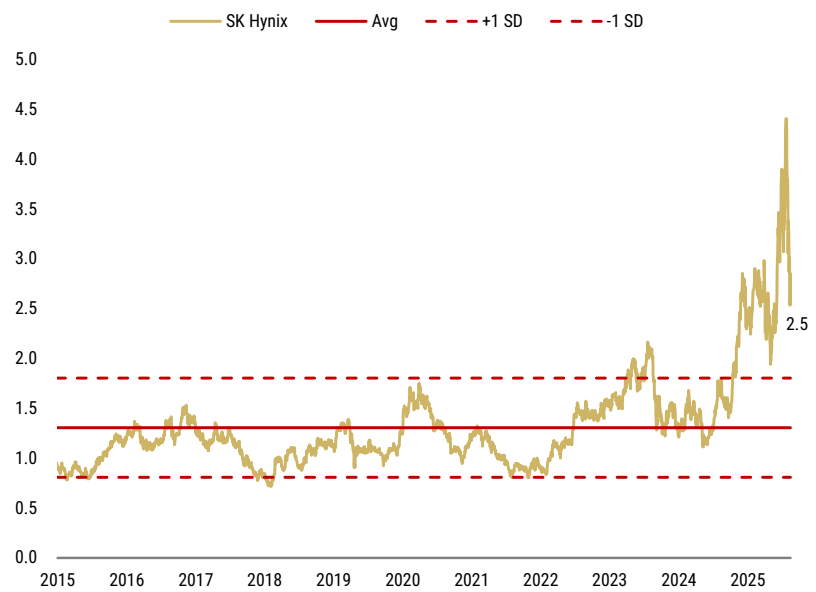
Valuations have pulled back sharply

P/B multiples have retreated from recent highs – Samsung to 1.7x and SK Hynix to 2.5x – yet both still sit above their long-run averages.

Samsung Electronics — P/B (x)



SK Hynix — P/B (x)



Source: Factset, Morgan Stanley Research estimates

CXMT: China’s DRAM champion lands a record ~\$85bn STAR Market IPO

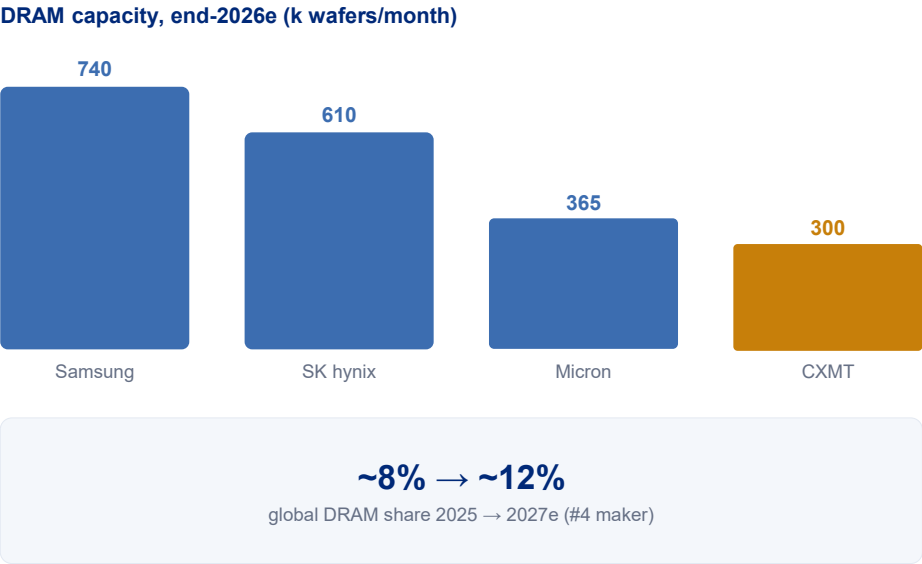
CXMT’s ~\$8.5bn Shanghai listing – the largest STAR Market IPO ever.

**Profile**

- Founded 2016 · HQ Hefei, Anhui
- China’s only large-scale DRAM maker
- Node: D1z (1z-nm) DDR5 / LPDDR5X, no EUV
- ~99% of 2025 revenue is DDR / LPDDR

**IPO – Shanghai STAR Market**

- Price ¥8.66/sh · trading from 27 Jul
- Raise ~¥57.9bn (~\$8.5bn); up to ~\$9.8bn
- Largest STAR Market IPO ever; Asia’s biggest 2026
- Implied value ~¥579bn (~\$85bn)
- FY26e P/E ~5.8x (trailing ~309x); P/B ~3x
- Strategic: Alibaba Cloud, Meituan, Xiaomi



CXMT fills the commodity DDR5 gap vacated by the Big-3 HBM pivot; the HBM gap remains the structural constraint.

Source: Trendforce, Company prospectus; Reuters; SCMP.

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**Daniel Yen**

Greater China Semiconductor Research Analyst

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Memory pricing is very high and bandwidth is limited

Exhibit 13: DDR5 16Gb 2Gx8 ASP evolution

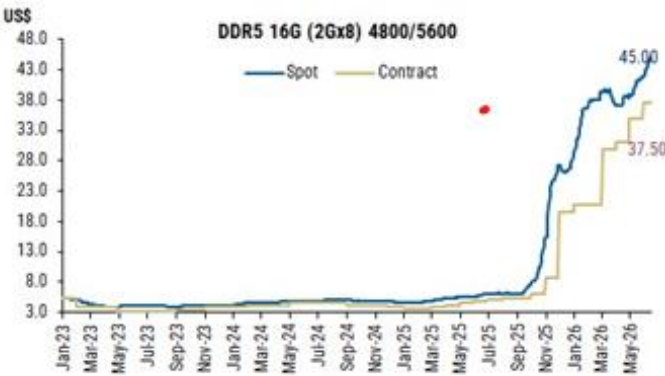
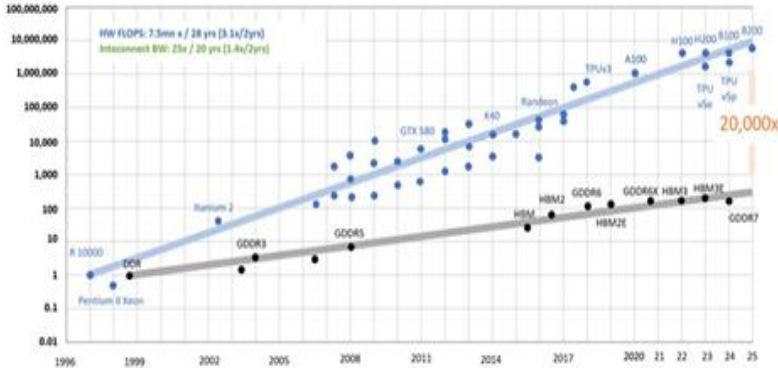


Exhibit 15: Compute vs. Memory comparison



Source: Morgan Stanley Research

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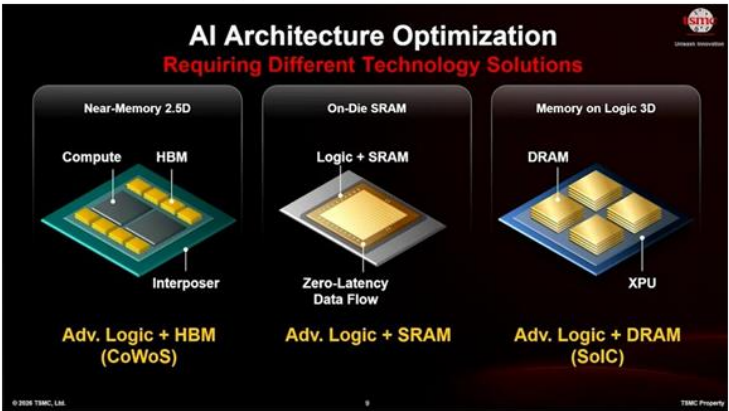
Example of memory innovation

Exhibit 41: Qualcomm's HBC Technology



Source: Qualcomm

Exhibit 42: AI Architecture Optimization



Source: TSMC

Six vectors to frame memory innovation

Exhibit 16: Summary of different paths of innovation



Source: Company data, Morgan Stanley Research estimates



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We estimate TAM to be around USD25bn

Exhibit 17: Memory Innovation TAM estimates - excluding HBM, base case (US\$m)

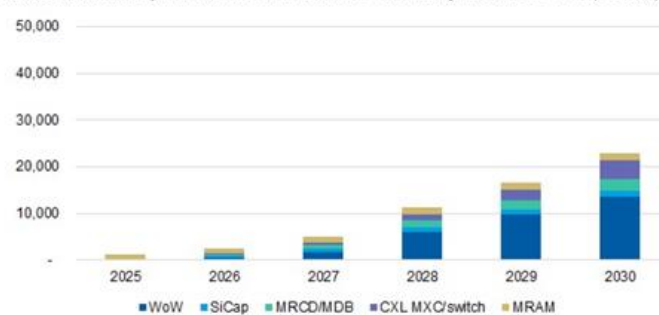
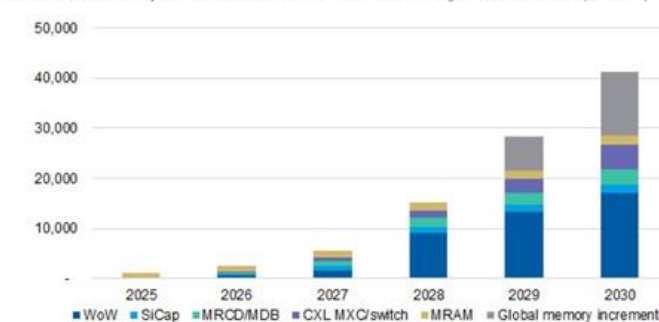


Exhibit 18: Memory Innovation TAM estimates - excluding HBM, bull case (US\$m)



Source: Company data, Morgan Stanley Research estimates

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Global memory comp

Exhibit 20: Global memory comparables table

| Ticker    | Company             | Price<br>7/19/2026 | Curr<br>ency | Price<br>Target | Upside/<br>Downside | Rating | Dividend<br>Yield<br>[%] | PDP<br>Yield<br>[%] | Market<br>Cap<br>(US\$M) | 3M Avg.<br>Daily<br>Trading<br>(US\$M) | P/E Ratio (x) |       |       | EPS Growth |       |       | ROAE  |       |       | P/B Ratio (x) |       |       |
|-----------|---------------------|--------------------|--------------|-----------------|---------------------|--------|--------------------------|---------------------|--------------------------|----------------------------------------|---------------|-------|-------|------------|-------|-------|-------|-------|-------|---------------|-------|-------|
|           |                     |                    |              |                 |                     |        |                          |                     |                          |                                        | 2026e         | 2027e | 2028e | 2026e      | 2027e | 2028e | 2026e | 2027e | 2028e | 2026e         | 2027e | 2028e |
| Memory    |                     |                    |              |                 |                     |        |                          |                     |                          |                                        |               |       |       |            |       |       |       |       |       |               |       |       |
| 005930.KS | Samsung Electronics | 285,000            | KRW          | 381,000         | 34%                 | O      | 1.4%                     | 2.9%                | 1,411,827                | 5,603.3                                | 5.9           | 4.3   | 4.3   | 722%       | 38%   | 0%    | 61%   | 50%   | 34%   | 2.5           | 1.8   | 1.1   |
| 000660.KS | SK hynix            | 2,180,000          | KRW          | 2,600,000       | 19%                 | O      | 0.5%                     | 7.6%                | 1,182,083                | 6,660.7                                | 6.8           | 4.7   | 4.4   | 437%       | 44%   | 9%    | 100%  | 85%   | 42%   | 4.6           | 2.3   | 1.5   |
| MIU.O     | Micron              | 979.3              | USD          | 1,250.0         | 23%                 | O      | 0.0%                     | NA                  | 1,138,198                | 41,306.3                               | 9.4           | 5.4   | NM    | 883%       | 75%   | NA    | 88%   | 89%   | 47%   | 8.4           | 3.9   | 2.8   |
| 285A.T    | Kioxia              | 77,000.0           | CNY          | 110,000.0       | 43%                 | O      | 0.0%                     | NA                  | 281,941                  | 21,475.6                               | 10.6          | 8.0   | 8.5   | 2259%      | 33%   | -6%   | 145%  | 73%   | 42%   | 8.1           | 4.4   | 3.2   |
| SNDK.O    | Sandisk             | 1,915.9            | USD          | 1,750.0         | -9%                 | O      | 0.0%                     | NA                  | 270,850                  | 20,568.8                               | 13.9          | 9.8   | NM    | 321%       | 41%   | NA    | 68%   | 83%   | 41%   | 14.5          | 5.3   | 3.5   |
| 603986.SS | Giga Device         | 612.0              | CNY          | 888.0           | 45%                 | O      | 0.3%                     | 4.1%                | 80,106                   | 3,570.1                                | 36.7          | 20.2  | 17.4  | 574%       | 81%   | 16%   | 47%   | 54%   | 43%   | 13.4          | 8.5   | 6.2   |
| 2408.TW   | Nanya Tech          | 435.5              | TWD          | 550.0           | 26%                 | O      | 0.0%                     | 13.6%               | 42,384                   | 1,290.7                                | 6.6           | 4.7   | 8.6   | 2677%      | 42%   | -45%  | 77%   | 57%   | 22%   | 3.5           | 2.9   | 1.6   |
| 301308.SZ | Longsys             | 587.6              | CNY          | 673.0           | 15%                 | E      | 0.1%                     | NA                  | 30,219                   | 1,544.0                                | 10.1          | 14.2  | 23.4  | 1585%      | -35%  | -39%  | 122%  | 43%   | 20%   | 7.9           | 5.1   | 4.3   |
| 2344.TW   | Winbond             | 176.5              | TWD          | 288.0           | 63%                 | O      | 0.0%                     | 18.6%               | 25,060                   | 967.5                                  | 8.3           | 4.6   | 4.1   | 2318%      | 80%   | 13%   | 61%   | 59%   | 41%   | 3.9           | 2.1   | 1.4   |
| 6299.TWD  | Phison              | 2,220.0            | TWD          | 2,588.0         | 17%                 | E      | 1.5%                     | 1.6%                | 15,279                   | 535.4                                  | 6.5           | 16.7  | 39.7  | 758%       | -61%  | -58%  | 79%   | 23%   | 10%   | 3.7           | 4.0   | 4.1   |
| 58MD.O    | Silicon Motion      | 326.4              | USD          | 400.0           | 23%                 | O      | 2.2%                     | 2.1%                | 11,068                   | 262.8                                  | 34.6          | 18.6  | 11.8  | 198%       | 86%   | 57%   | 33%   | 45%   | 49%   | 10.0          | 7.1   | 4.9   |
| 688110.SS | Daxion              | 164.0              | CNY          | 20.0            | -88%                | U      | 0.0%                     | 0.6%                | 10,567                   | 514.8                                  | NM            | NM    | NM    | -102%      | 1266% | 176%  | 0%    | 2%    | 6%    | 23.9          | 23.5  | 22.5  |
| 2337.TW   | Macronix            | 143.5              | TWD          | 220.0           | 53%                 | O      | 4.6%                     | 17.2%               | 8,407                    | 679.6                                  | 14.6          | 5.7   | 5.5   | -652%      | 154%  | 3%    | 36%   | 57%   | 37%   | 4.3           | 2.4   | 1.6   |
| Mean:     |                     |                    |              |                 |                     |        | 6.6%                     | 7.4%                |                          |                                        | 13.7          | 9.8   | 12.8  | 918%       | 202%  | 15%   | 57%   | 43%   | 28%   | 6.8           | 6.8   | 5.8   |
| Median:   |                     |                    |              |                 |                     |        | 6.1%                     | 4.1%                |                          |                                        | 9.8           | 6.9   | 8.5   | 666%       | 81%   | 8%    | 54%   | 48%   | 23%   | 6.1           | 6.8   | 4.2   |

E = Morgan Stanley Research estimates. Source: Company data, Morgan Stanley Research

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Europe Semiconductor Research Associate

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ASML NDR Takeaways: Signals Long-Term Growth Drivers

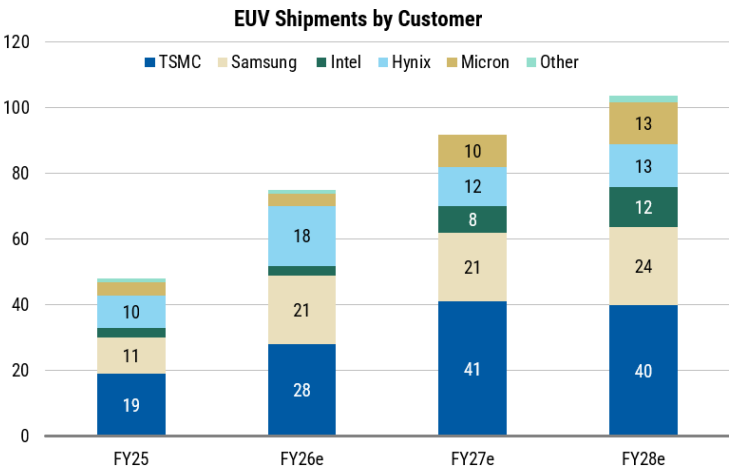


Source: Morgan Stanley Research

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ASML: We see 92 and 104 EUV Shipments for FY27 and FY28 respectively



Source: Morgan Stanley Research estimates

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US Technology Hardware Research Associate



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| Overweight/Buy        | 1544              | 42%        | 453                              | 49%            | 29%                  | 757                                               | 44%                   |
| Equal-weight/Hold     | 1577              | 43%        | 390                              | 42%            | 25%                  | 769                                               | 44%                   |
| Not-Rated/Hold        | 3                 | 0%         | 1                                | 0%             | 33%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 544               | 15%        | 89                               | 10%            | 16%                  | 204                                               | 12%                   |
| Total                 | 3,668             |            | 933                              |                |                      | 1731                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

## Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

## Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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## INDUSTRY COVERAGE: Technology - European Semiconductors

| COMPANY (TICKER)                         | RATING (AS OF) | PRICE* (07/20/2026) |
|------------------------------------------|----------------|---------------------|
| <b>Lee Simpson</b>                       |                |                     |
| ASML Holding NV (ASML.AS)                | O (09/22/2025) | €1,520.80           |
| Infineon Technologies AG (IFXGn.DE)      | O (02/06/2025) | €64.00              |
| STMicroelectronics NV (STMPA.PA)         | O (03/26/2026) | €54.91              |
| <b>Nigel van Putten</b>                  |                |                     |
| Aixtron SE (AIXGn.DE)                    | E (05/25/2023) | €39.19              |
| ASM International NV (ASML.AS)           | O (06/19/2024) | €856.80             |
| BE Semiconductor Industries NV (BESI.AS) | O (11/07/2022) | €223.50             |
| Melexis N.V. (MLXS.BR)                   | E (02/05/2026) | €69.05              |
| Nordic Semiconductor ASA (NOD.OL)        | E (02/10/2025) | NKr 157.70          |
| Soitec SA (SOIT.PA)                      | O (03/26/2026) | €86.40              |
| VAT Group AG (VACN.S)                    | E (03/21/2025) | SFr 666.60          |

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\* Historical prices are not split adjusted.